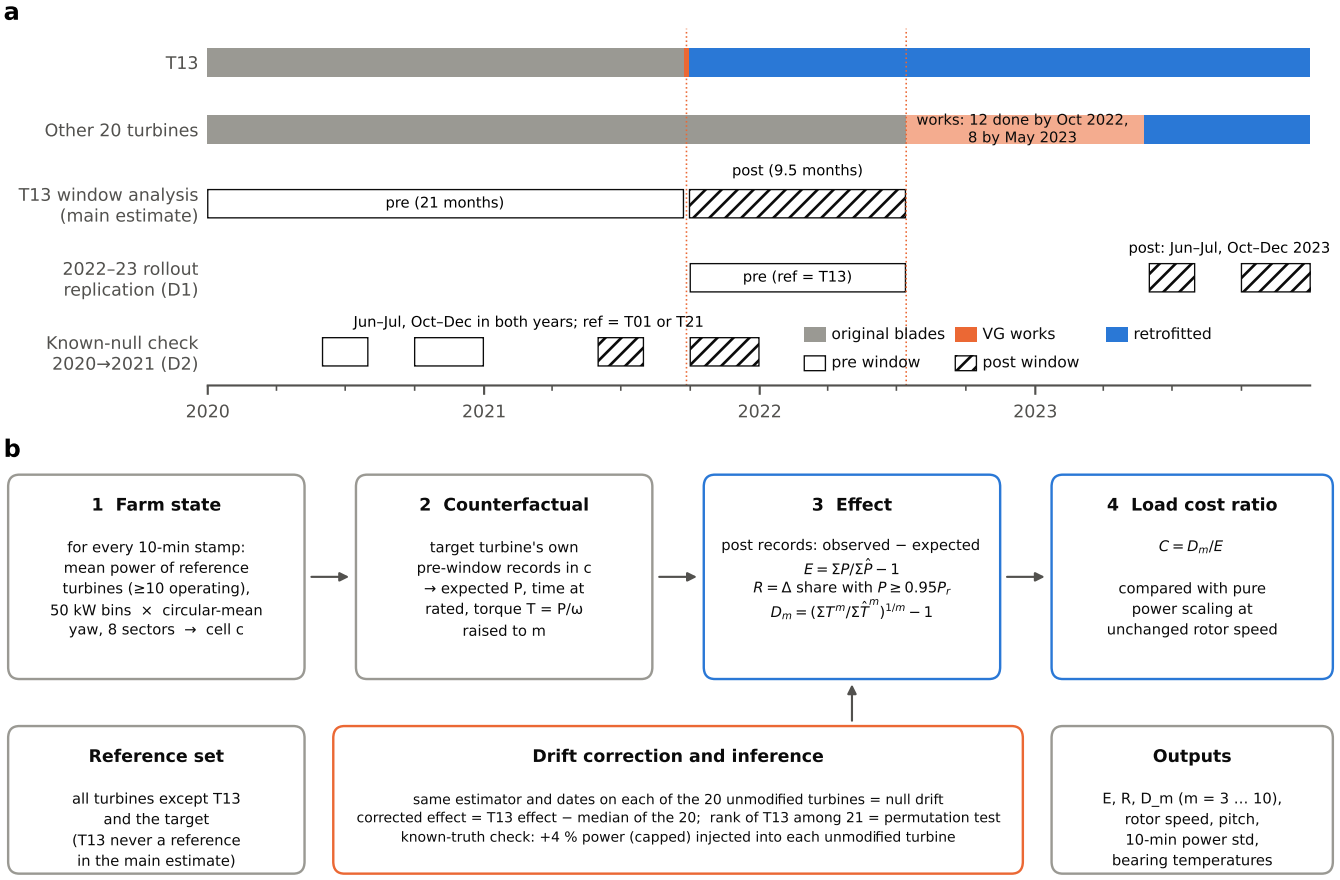
a
T13
Other 20 turbines
works: 12 done by Oct 2022,
8 by May 2023
T13 window analysis
(main estimate)
post (9.5 months)
pre (21 months)
2022–23 rollout
replication (D1)
pre (ref = T13)
post: Jun–Jul, Oct–Dec 2023
Known-null check
2020→2021 (D2)
Jun–Jul, Oct–Dec in both years; ref = T01 or T21
original blades
VG works
retrofitted
pre window
post window
2020
2021
2022
2023
b
1 Farm state
for every 10-min stamp:
mean power of reference
turbines (≥10 operating),
50 kW bins × circular-mean
yaw, 8 sectors → cell c
2 Counterfactual
target turbine's own
pre-window records in c
→ expected P, time at
rated, torque T = P/ω
raised to m
3 Effect
post records: observed − expected
$E = \Sigma P / \Sigma \hat{P} - 1$
$R = \Delta$ share with $P \geq 0.95 P_r$
$D_m = (\Sigma T^m / \Sigma \hat{T}^m)^{1/m} - 1$
4 Load cost ratio
$C = D_m / E$
compared with pure
power scaling at
unchanged rotor speed
Reference set
all turbines except T13
and the target
(T13 never a reference
in the main estimate)
Drift correction and inference
same estimator and dates on each of the 20 unmodified turbines = null drift
corrected effect = T13 effect − median of the 20; rank of T13 among 21 = permutation test
known-truth check: +4 % power (capped) injected into each unmodified turbine
Outputs
E, R, D_m (m = 3 … 10),
rotor speed, pitch,
10-min power std,
bearing temperatures